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FINAL PROJECT REPORT
SMART FUNERAL APPLICATION

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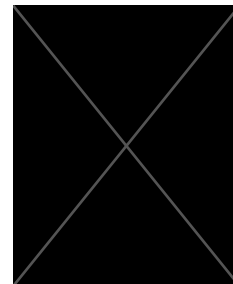


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1.0 Introduction

Smart Funeral Application is an advanced approach to the ever-evolving technology mainly catering the ease of planning and coordinating funeral arrangements for both parties, grieving families and the funeral service providers. The application brings upon an opportunity to plan funeral arrangements with a hint of digital platforms which focuses on the features with several functions which integrate to provide a well performing application. The groundbreaking application offers both practical assistance in planning and coordinating funeral arrangements and a personalized approach.

1.1 Problem Statement

With the evolving technological aspects and societal transformation, funeral arrangements can be troublesome for some because it involves different stages of planning and coordination. One of the pivotal problems faced by many is due to the disintegrated communication and coordination. Apart from that, time consuming administrative tasks in planning a funeral arrangement is also one of the problems faced by the society. Besides, the lack of accessibility for all demographics is also one of the key problems.

1.2 Justification of the Problems Potentially to be Solved

One of the problems stated is the disintegration of communication and coordination creates stress and confusion among grieving families. In the absence of a centralized digital platform, many discussions may be overlooked leading to misunderstandings. Besides, time consuming administrative tasks often involves a lot of paperwork and documentation which may create stress and delays among them. Moreover, the other problem being lack of accessibility for all demographics excludes those with varying levels of technological proficiency from attending the funeral event.

1.3 Target Users

The target users for the app may comprehend a diverse range of individuals. The primary users of the app are the grieving family, relatives, friends and funeral service providers. These are the primary users of the application as they take the initiative in planning to coordinate a funeral for the deceased ones. Individuals planning in advance are also one of the key users of the Smart

Funeral Application. As they are proactively planning their own funeral arrangements in advance, it will be useful for their family members to do the arrangements as per their preference.

1.4 Challenges that can be Helped

The development and implementation of the Smart Funeral Application has all the potential to solve the problems faced by society these days. For example, the challenges that can be helped by the application is that it connects people from different demographic areas to join in an online meeting to prepare the funeral arrangements. The development of the application also documents and updates about the deceased ones. Family members are notified when there is any event. The features built in the app are beneficial in planning and coordinating a funeral event.

1.5 Related Work and Existing Technologies

As per the database, one of the few similar works that is found online is that James Vlahos, the director of conversational AI programmed a chatbot which is capable of reviving the deceased ones' stories via text messages, audio, images and videos that created an interactive experience with the deceased ones. He then launched HereAfter AI which is a US-based company where people are allowed to upload their photos and memories that are turned into a life-story avatar with whom they can communicate.

2.0 Final Design Overview

2.1 Purpose of the Application

With features like digital memorials, legacy videos, and event planning, the funeral app provides users with a comprehensive platform to help with the memorialization process. A sense of control over digital legacies is fostered by user-centric design, which guarantees safe access to personalized profiles. The goal of the app is to be a sympathetic companion for celebrating and remembering loved ones.

2.2 Supported Features

2.2.1 User Authentication

With features like digital memorials, legacy videos, and event planning, the funeral app provides users with a comprehensive platform to help with the memorialization process. A sense of control over digital legacies is fostered by user-centric design, which guarantees safe access to personalized profiles. The goal of the app is to be a sympathetic companion for celebrating and remembering loved ones.

2.2.2 Profile Management

The funeral app's Profile Management feature provides a methodical way for users to make and manage profiles, including those of the deceased. Users have the option to regulate privacy settings, share narratives, upload images, and customize details. This tool blends IoT and AR to create a dynamic environment that improves personalisation and allows for the addition of memories. User-centric and adaptive profile management is made possible via IoT connectivity.

2.2.3 Legacy Video Messages

Using the funeral app, users can record video messages and write letters to express enduring thoughts using the Legacy Video Messages feature. Messages can be scheduled by users, guaranteeing a permanent record timed to coincide with important occasions. AI analysis generates emotionally relevant answers by interpreting shared content through NLP. Adaptive

learning allows the AI to send more appropriate and customized condolence messages over time, corresponding with various grieving styles.

2.2.4 Digital Memorialization

The funeral app's Digital Memorialization feature creates a special area where users may honor deceased loved ones virtually. This involves creating a possible online memorial wall that would serve as a platform for condolence messages and promote harmony. Customized avatars are a feature of augmented reality that provide a virtual sense of friendship. The AR Memorials promote teamwork by enabling friends and family from all over the world to participate at the same time and leave upgraded messages on a multimedia-rich, spatialized memorial wall.

2.2.5 Event Planning

The funeral app's Event Planning feature simplifies the planning of memorial events by using calendars, polls, and forms to choose dates and times that work best for attendees. Participant availability is gathered via forms, guaranteeing ideal scheduling. Quick decision-making is facilitated by polls, and event schedules are visualized using calendars. Notifications and reminders improve user participation. By taking into account variables like desired times and historical attendance trends, AI-driven scheduling proactively suggests times that are in line with participant availability, guaranteeing relevant events that are well-attended.

2.2.6 Yearly Anniversaries

The funeral app's Yearly Anniversaries feature gives users the ability to remember loved ones by creating customized reminders. In an effort to promote unity, users can virtually light candles, send flowers, and take part in online commemoration events on the anniversary. Through the addition of virtual reality (VR) event rooms to the Event Planning tool, the app expands the user experience. These immersive environments, which can be altered to suit the atmosphere of the event, enable emotionally stirring conversations and connect geographically separated people.

2.3 Functional Requirements

2.3.1 User Authentication

Acceptance Criteria:

- The system shall validate the recognised fingerprint or facial identification against the stored credentials.
- The system shall implement Two Factor Authorisation through email or SMS.
- A user shall resort to logging in through username and password if two consecutive unsuccessful biometric login attempts occur.
- A user account shall be locked if three consecutive unsuccessful login attempts occur.
- The account lockout duration shall be 30 minutes.
- Successful authentication shall grant the user access to the system's features.
- The system shall validate the entered username and password against the stored credentials.

2.3.2 Profile Management

Acceptance Criteria:

- The system shall allow users to upload pictures,edit personal details.
- The system shall allow users to control the visibility and accessibility of the profiles
- The system shall include holographic displays of the user's profile elements.
- Users shall be able to create new and manage profiles.

2.3.3 Legacy Video Messages

Acceptance Criteria:

- Users shall be able to write farewell messages and letters.
- Users shall be able to control which media can be viewed by the public or has restricted view.
- The system shall allow users to schedule the release of the video messages according to time and date.
- The system shall analyze user shared text and media content to understand users behavior and attributes.

- The system shall be able to generate condolence and closure messages mimicking users texting and speech behavior.
- Users shall be able to record and upload video messages.

2.3.4 Digital Memorialization

Acceptance Criteria:

- The system shall create a virtual memorial wall.
- The system shall allow other users to upload memories in the virtual memorial wall.
- The system shall implement personalized avatars.
- The system shall provide users with a dedicated space for digital memorialization.

2.3.5 Event Planning

Acceptance Criteria:

- Users shall be able to efficiently plan memorial events using forms, polls, and calendars.
- The system shall include reminders and notifications to inform attendees about event details and any changes.
- The system shall include AI-driven scheduling and should optimize event attendance based on participant preferences.

2.3.6 Yearly Anniversaries

Acceptance Criteria:

- Users shall have the ability to set personalized reminders and notifications for significant yearly anniversaries.
- The system shall have virtual features like candle lighting, sending virtual flowers
- The system shall have online video meeting space.
- The system shall include VR event spaces for immersive virtual memorial events.
- The system shall allow the VR event space to be customized based on the event's mood and importance.

2.4 Non-functional Requirements

2.4.1 Security

- The user authentication process such as the Two Factor Authentication and Biometric Recognition must comply with industry security standards.
- User data such as personal information, videos, messages must be encrypted and stored securely.
- Security testing and audits must be carried out regularly to identify and address potential security vulnerabilities.

2.4.2 Reliability

- The system must have high availability, minimizing downtime for users to access the application's features.
- The system must have backup and recovery mechanisms to protect against data loss.

2.4.3 Usability

- The user interface must be user friendly and intuitive.
- The user interface must provide a seamless user experience.
- The system must include accessibility features.

2.4.4 Compatibility

- The system must be compatible with all ranges of devices including versions of IOS, Windows, Android and so on.
- The Virtual and Augmented Reality features must be compatible with available AR and VR devices.

2.4.5 Adaptability

- The app must be adaptable to technological advancements.
- The app must allow integration of new emerging technologies.
- The app must have regular updates and maintenance.

2.4.6 Performance

- The system must provide fast and responsive authentication, profile management, and content uploading, ensuring a seamless user experience.
- The system response times for AI-generated messages and recommendations must be within acceptable limits to maintain user engagement.
- Virtual and augmented reality features must deliver smooth interactions and load times for an immersive experience.

2.5 Design Evolution

2.5.1 Loading Screen

Initial Design

- The initial design did not contain a loading screen.
- There might be a delay in content loading leading to a potential user experience issue.
- There is a need for a loading screen to provide visual feedback
- The loading screen is needed to prevent a disorienting experience for users.

Evolution and Reasoning

- A loading screen has been implemented to enhance user awareness during data retrieval.
- The loading screen ensures a smoother user experience, even in scenarios with longer loading times.

2.5.2 Main Menu

Initial Design

- The initial design did not contain any central navigation system.
- This might potentially cause user navigation challenges.
- App navigation system is improved by introducing a dedicated main menu.

Evolution and Reasoning

- The main menu serves as a centralized hub for navigation, providing users with easy access to various app features.

- The categorized menu structure was implemented to improve user understanding and streamline the overall navigation experience.

2.5.3 Sign Up Method

Initial Prototype

- Multiple signup methods were offered including Google, Apple, and others, providing users with various registration choices.
- This might cause potential confusion and complexity in the signup process due to multiple options, which could impact user onboarding.

Evolution and Reasoning

- The signup was changed to only email registration, simplifying the onboarding process for users and reducing decision fatigue.
- The external signup options were removed to focus on a more straightforward and consistent user experience.
- The email only sign up also decreases the surface area of security vulnerabilities.

2.5.4 New Account Information

Initial Prototype

- The initial sign up process collected a comprehensive set of user information including details like date of birth.
- User privacy had to be prioritized during the initial sign up process.

Evolution and Reasoning

- The signup process was modified to collect basic information (name,email, password) providing a more user-friendly and privacy-conscious experience.
- Profile management was implemented for users to edit and add additional information at their convenience, ensuring flexibility without overwhelming users during the initial sign up.

2.5.5 Profile Management

Initial Design

- The initial design integrated the user profile within the feed, combining user-specific information such as the profile picture and about section with the main content feed.
- The initial design also included a search option within the feed for users to search for others.
- The initial design also did not include a separate interface for settings.
- The initial design did not include a separate interface for the user's own profile view and a feed with updates from other users.

Evolution and Reasoning

- The evolved design opted to separate the user profile from the feed, providing distinct interfaces for these two core features. This separation enhances clarity and focuses on providing a dedicated space for personal information and social content.
- Users can now navigate their profile independently without the distraction of the main feed, allowing for a more immersive and focused user experience.
- A settings interface was introduced, ensuring users can manage account preferences independently.
- Users can now view updates from other users in a dedicated feed interface, allowing for more straightforward and immersive content consumption.

2.5.6 Legacy Video Messages

Initial Design

- The initial design included all of the features such as gallery of videos, letters, chatbox into 1 interface.

Evolution and Reasoning

- The single interface is separated into two, making it less congested and easier on the eyes of the user.
- One of the interfaces includes a place to upload and edit, record the legacy messages.
- The other interface includes the setting of privacy where the user can choose who can view their legacy messages and also set time and date to release the legacy messages.

2.5.7 Event Planning

Initial Design

- The initial design has only one interface.

- The initial interface contains a calendar, and a media gallery segregated according to the years.
- The initial design also has a chatbox and form and poll section for the participants of the event to decide on a date.

Evolution and Reasoning

- Individual interface is added for the users to choose between available online meeting according to their needs.
- A separate interface for chatbox and polls for fellow participants in the event to discuss.

2.5.8 Digital Memorialization

Initial Design

- The digital memorialization interface is a single interface containing a chat box for other users to share stories and experiences, good memories about the deceased user.
- This interface also includes a section for other users to upload videos and pictures about the deceased user to share their memories.

Evolution and Reasoning

- The interface is edited to be less congested but follows the initial interface design.

2.5.9 Yearly Anniversary

Initial Design

- The initial design includes a flowers column where the guest for the anniversary can give virtual flowers for the deceased ones.
- The initial design also includes a commemoration box which allows the guests to celebrate the deceased ones by writing messages about them.
- The initial design includes a meeting link for the yearly anniversary and VR effects that can be added to the meeting.

Evolution and Reasoning

- There is a timeline of the user's timeline added to the interface.
- There is a separate interface for the users to choose between gifts and flowers to be given to the deceased user.
- Also the gifts, flowers options have been increased.

2.6 Design Decisions

2.6.1 Color Scheme

Navy Blue and Pastel Yellow

- The color yellow is a symbol of coziness, optimism, and hope. Within the funeral app context, it offers a consoling and uplifted feeling, assisting users in navigating a trying time with a sense of peace.
- The color navy blue stands for depth, stability, and trust. Navy blue gives the app's visual identity a sense of sophistication and serenity that gives users confidence and trust at delicate times.

2.6.2 Name of the Application

Eternal Ease

- Eternal captures the enduring quality of emotions and memories. Users are reassured that their memories are honored and preserved by its sense of permanence.
- Ease connotes a cozy and user-friendly environment. The app seeks to make memorialization, funeral planning, and grieving process navigation easier.

2.6.3 Application Logo

Phoenix

- The phoenix represents tenacity, rebirth, and everlasting life. It suggests that memories can rise and endure, much like the fabled phoenix that rises from its ashes, which is consistent with the app's theme of everlasting memories.

2.6.4 Typography

Canva Sans

- Canva Sans is a contemporary sans-serif font that is easy to read and has a sleek appearance. It ensures that information is presented clearly and sympathetically while also adding a professional and reassuring tone that blends in well with the app's design.

In conclusion, the Eternal Ease funeral app's design choices seek to provide users with a calming, emotionally comforting environment.

3.0 Implementation

3.1 Internal Implementation

The app places a high priority on internal security by utilizing strong user authentication, industry-accepted encryption techniques, and safe data storage methods. Legacy video messaging uses end-to-end encryption to guarantee private conversation. Secure methods are leveraged by intuitive profile management to provide individualized experiences. The framework for digital memorialization combines careful data handling with multimedia components. Real-time notifications, site mapping, scheduling algorithms, and an intelligent system for yearly anniversary reminders are all part of simplified event preparation on the inside.

3.2 Design Decisions

The Smart Funeral App prioritizes a seamless user experience, emphasizing security in user authentication with industry-standard protocols and potential biometric integration. Intuitive interfaces across profile management, legacy video messages, and digital memorialization enhance usability. Data protection is paramount. The collaborative event planning platform and customizable memorial options ensure emotional and practical support, embodying a compassionate interface for funeral-related tasks. The overall goal of these design decisions is to provide users navigating the delicate environment of funeral-related tasks with an interface that is emotionally resonant, safe, and supportive.

3.3 Implementation Usability Impact

The Smart Funeral App creates a user experience based on deliberate design decisions by giving emotional support, security, and simplicity top priority. Robust user authentication fosters confidence. Viewing video chats and managing profiles are made easier with intuitive UI. Digital memorialization with multimedia strengthens emotional ties. Coordination is made easier by simplified event preparation, and users may navigate funeral-related duties in a sympathetic online environment thanks to customisable memorials that include automated anniversary reminders.

3.4 User Test Evaluation Method

- User Authentication Testing

Objective	Evaluate the usability and security of the user authentication system.
Tasks	1. Log in to the app 2. Attempt a password reset and assess the clarity of the process
Completion Status	Complete
Result	Objective achieved

- Profile Management Evaluation

Objective	Evaluate the profile management section's privacy and usability.
Tasks	1. Update personal information and preferences 2. Adjust privacy settings and explore account customization options 3. View other user's profile
Completion Status	Complete
Result	Objective achieved

- Legacy Video Messages Testing

Objective	Evaluate the simplicity of recording, editing, and sharing legacy video messages.
Tasks	1. Record a video message 2. Share the video message with a designated recipient 3. View video messages from deceased family
Completion Status	Complete
Result	Objective achieved

- Digital Memorialization Testing

Objective	Evaluate how well the interface support personalization and the smoothness of sending gifts to deceased acquaintances
Tasks	1. Create a digital memorial for a loved one

	2. Add multimedia elements such as photos, messages, and anecdotes 3. Send gifts to loved ones
Completion Status	Complete
Result	Objective achieved

- Event Planning Usability Assessment

Objective	Evaluate the collaborative and accessible aspects of the event planning feature
Tasks	1. Schedule a fictional funeral event 2. Explore location mapping features
Completion Status	Complete
Result	Objective achieved

- Yearly Anniversaries Testing

Objective	Assess the effectiveness of automated reminders and customization options for commemorations
Tasks	1. Interact with automated yearly anniversary reminders 2. Customize commemoration options
Completion Status	Complete
Result	Objective achieved

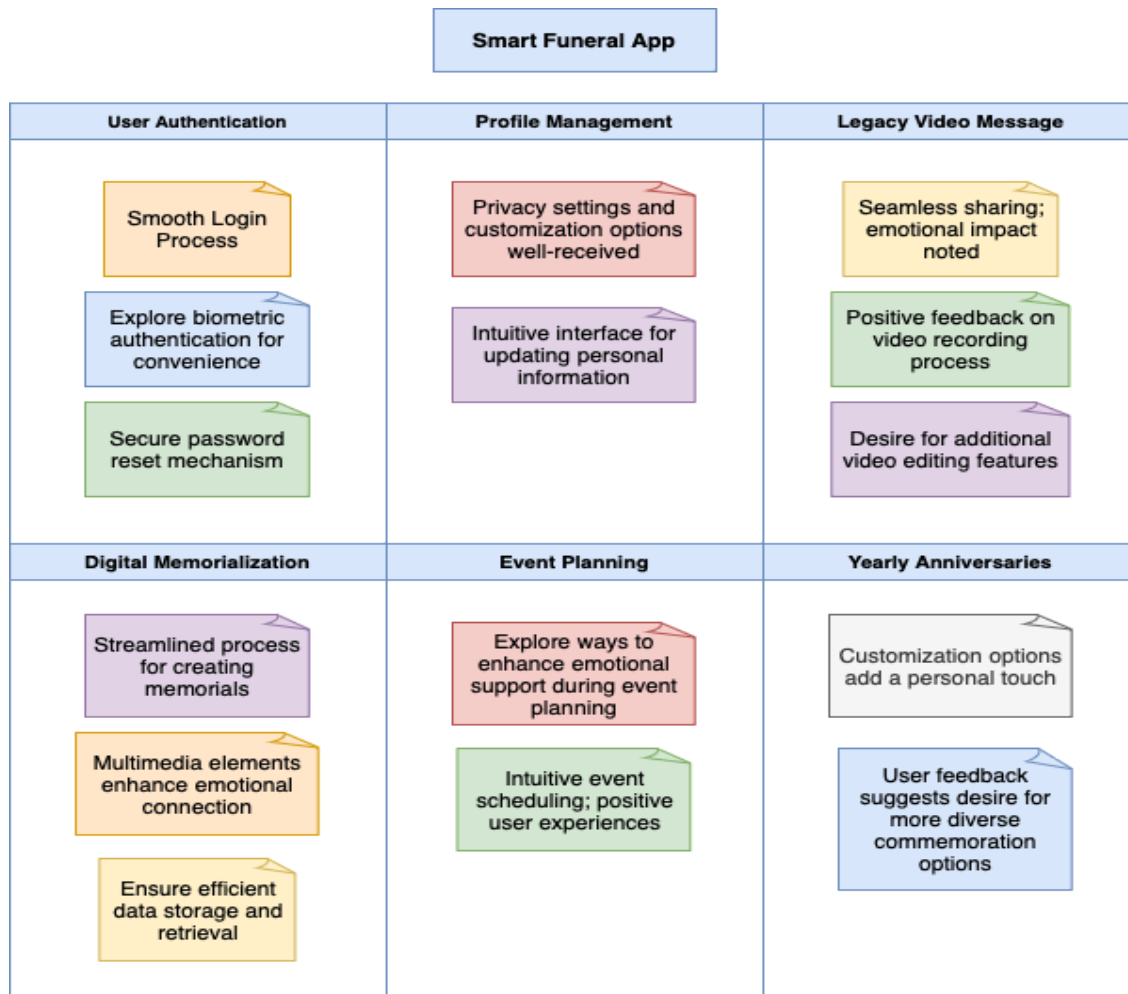
3.5 Users Tested Details

Number of Users: 30 individuals involved

User Characteristics: Participants were selected to represent the primary user group of the smart funeral app, including individuals aged 30-60, who may have experienced funeral-related activities or are anticipating the need for such services. The participants were diverse in terms of age, gender, and cultural background to guarantee a thorough assessment of the emotional resonance and usability of the app.

Recruitment Approach: Participants were personally emailed and messaged to get their permission to participate in this user testing phase. A call for volunteers was also posted on social media. In order to make sure that participants had suitable expectations or experiences with funeral-related activities, screening questions were included into the recruitment process.

3.6 Affinity Diagram



4.0 Evaluation Results

4.1 Affinity Diagram Description

The app has features like user authentication, profile management, digital memorialization, legacy video messages, event planning, and yearly anniversaries that are intended to assist people in organizing and managing their funerals. User authentication, which enables users to register for an account and access the app, is where the diagram begins. Users can create a profile and add details like name, birthdate, and contact details once they have logged in. They have the option to add movies and images to their profile.

The next stage, known as digital memorialization, enables people to make a digital memorial that their loved ones can view. A memorial may contain stories, films, pictures, and other artifacts. It is also an option for users to share their memorial with others. The next step is event planning, which enables users to schedule their memorial service or funeral. Along with choosing the music, readings, and other elements, users can also customize the service's date, time, and location. The final step is yearly anniversaries, which enables users to schedule reminders for significant anniversaries like their own death date or a loved one's birthday. On these anniversaries, users have the option of sending emails or messages to their loved ones. In general, the Smart Funeral App is a feature-rich resource that assists individuals in organizing and coordinating their funerals. It has several features that can simplify and deepen the experience for the user and their loved ones.

4.2 Proof of Requirements Fulfillment

By offering a strong and secure user authentication system with biometric identification, Two Factor Authorization, and account lockout procedures, the smart funeral app effectively fulfills its functional requirements. Users can easily post photos, change personal information, and manage their profile's visibility and accessibility with the profile management function. Users can compose farewell messages, schedule releases, and even generate sympathy letters depending on their behavior, thanks to the Legacy Video Messages technology. Digital memorialization is achieved by putting together a personalized avatar and a virtual memorial wall where users and others can submit memories. AI-driven scheduling and reminders are added to the forms, polls, and calendars that make up the Event preparation tool, which

facilitates effective preparation of memorial events. Yearly Anniversaries are celebrated with customized alerts, virtual elements such as sending virtual flowers and lighting candles, and even immersive virtual reality (VR) event settings that may be altered according to the significance and mood of the occasion. The application satisfies its non-functional requirements as well, guaranteeing industry-standard security protocols, excellent dependability with backup and recovery systems, user-friendly and intuitive interfaces, compatibility with a wide range of devices, evolution with technology, and peak performance with respect to response times and seamless interactions for both VR features and AI-generated messages. To provide a seamless and safe user experience, regular security testing, compatibility checks, and upgrades are carried out.

4.3 System Success Analysis

Aspect	Description
Purpose	Alleviate complexities of funeral arrangements and memorialization
Target Users	Grieving families, relatives, friends, funeral service providers
Supported Features	User Authentication, Profile Management, Legacy Video Messages, Digital Memorialization, Event Planning, Yearly Anniversaries
Functional Requirements	User authentication (secure login, password reset, biometric recognition), Profile management (profile creation, editing, privacy control, holographic displays), Legacy video messages (recording, editing, scheduling, content filtering, AI-generated messages), Digital memorialization (virtual memorial wall, multimedia sharing, user-controlled visibility), Event planning (forms, polls, calendars, reminders, AI-driven scheduling), Yearly anniversaries (reminders, virtual features, VR event spaces)
Non-Functional Requirements	Security (industry-standard protocols, user data encryption, security testing), Reliability (high availability, data backup and recovery), Usability (user-friendly interface, accessibility features), Compatibility (support for various devices and operating systems), Adaptability (regular updates, integration of

	new technologies), Performance (fast response times, seamless user experience)
Design Evolution	Simplified sign-up process using email registration, Distinct interfaces for user profiles and the main feed, Enhanced customization options for profiles and privacy settings, Incorporation of AR and VR enhancements for immersive experiences, Rigorous security measures for user authentication and data protection
Implementation	Industry-standard protocols and secure storage solutions, User authentication with OAuth or JWT, Granular privacy control mechanisms in profile management
User Testing	User authentication (efficient login, secure password reset), Profile management (intuitive interfaces, customization), Legacy video messages (recording ease, smooth sharing), Digital memorialization (multimedia elements, privacy control), Event planning (form handling, poll execution, calendar updates), Yearly anniversaries (virtual features, VR event interactions)
Reflections	Effective in streamlining complexities, fostering emotional support, and facilitating memorialization
Conclusions	Significant contribution to funeral services, empowering families and professionals

Overall, the Smart Funeral Application appears to be a successful system that meets the needs of its target users. It has a wide range of features that are designed to make the funeral planning process easier and more efficient. The system is also secure and reliable, and it has been well-tested by users.

Here are some of the strengths of the Smart Funeral Application:

- It is a comprehensive system that addresses all of the needs of the funeral planning process.
- It is easy to use and navigate, even for people who are not tech-savvy.
- It is secure and reliable, and it protects the privacy of its users.
- It has been well-tested by users and has received positive feedback.

4.4 Root Cause Analysis

Even though the Smart Funeral Application has achieved a lot, there may be some problems that need to be looked at through Root Cause Analysis (RCA). The following are some possible issues and the underlying reasons behind them:

Issue: Poor uptake of users:

Root Cause 1: Ignorance: It is possible that the intended audience is ignorant of the existence or advantages of the application.

Root Cause 2: Marketing gaps: Ineffective communication channels or a failure to connect with the target audience are possible reasons why marketing strategies fail.

Root Cause 3: Accessibility problems: Due to technological constraints or a lack of language support, some user groups may encounter difficulties.

Issue: Excessive user attrition

Root Cause 1: A bad onboarding experience: Users may find it difficult or confusing to use the initial setup or feature navigation.

Root Cause 2: Unmet user expectations: It is possible that users' needs are not fully met by the application or that it does not provide features they value highly.

Root Cause 3: Technical problems: Users may experience performance problems, crashes, or bugs that make their experience less enjoyable.

Issue: Unexpected consequences or results

First root cause: Unexpected user behavior: Users may interact with the application in ways that were not considered during development, which could have unexpected results.

Root Cause 2: External factors: The efficacy of the application may be impacted by modifications to laws, regulations, or societal norms.

Root Cause 3: Ethical considerations: Reevaluating features may be necessary to account for unintentional biases or potential privacy concerns.

4.5 Possible Solutions

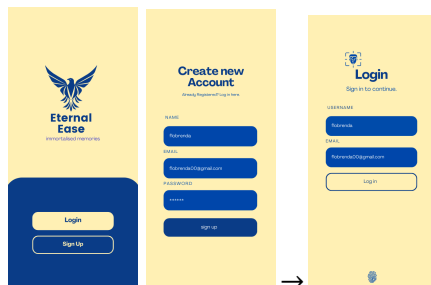
A diversified approach is necessary to increase awareness and user involvement in an era where digital solutions predominate. Using social media, partnerships, and community outreach in

focused marketing initiatives increases visibility. Optimizing app store listings and website SEO simultaneously improve discoverability. A dedication to diversity can be seen in addressing accessibility issues via assistive technology compatibility, a variety of language options, and unambiguous usage instructions. Analyzing comments and reports from users is very helpful in identifying and fixing problems. Iterative changes are prompted by design faults that are shown by usability testing. Ongoing improvement is ensured by remaining aware of user behavior, new trends, and outside factors. A defense against prejudice and privacy issues is the use of industry specialists in conjunction with ethical evaluations. Personalized recommendations and incentives along with strategic partnerships round out the user experience and make for a comprehensive and powerful application.

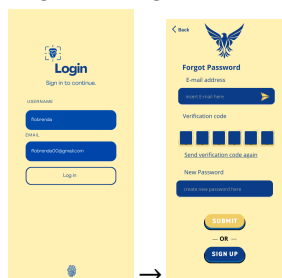
5.0 Reflections

5.1 Iterative Design Process

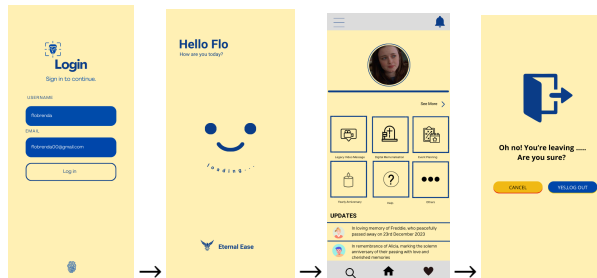
Sign up → login



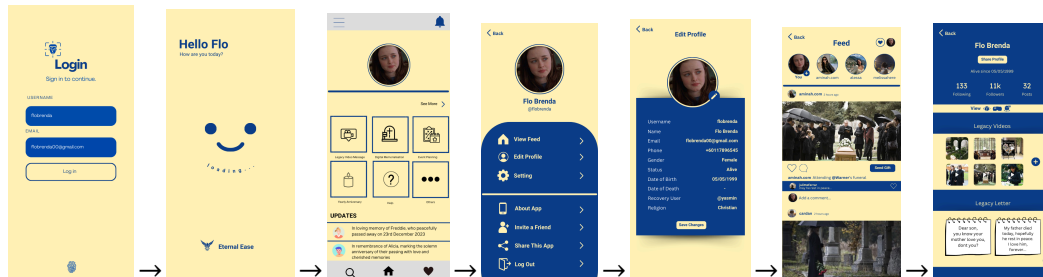
Login → Forgot password



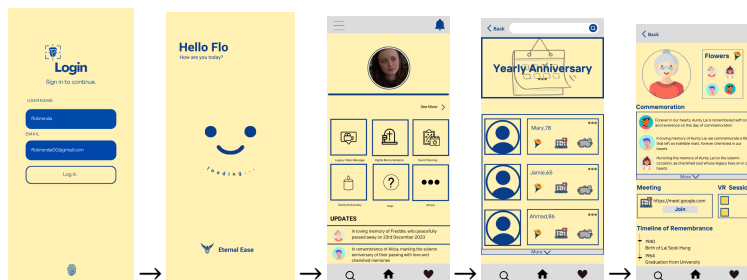
Login→Loading→Main menu→Log out



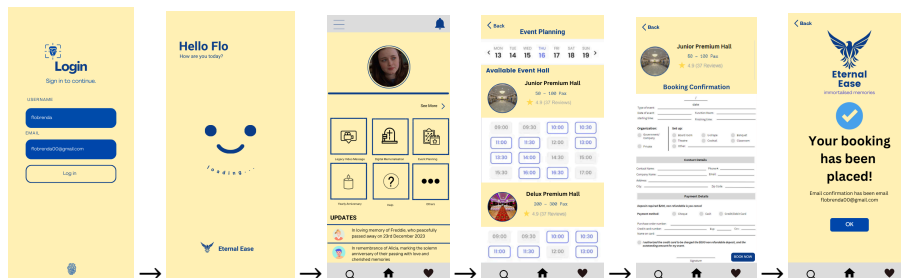
Login→Loading→Main menu →Profile management



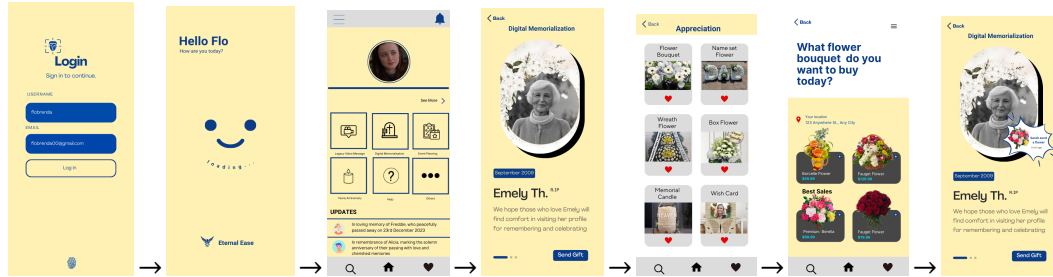
Login→Loading→Main menu→Yearly anniversaries



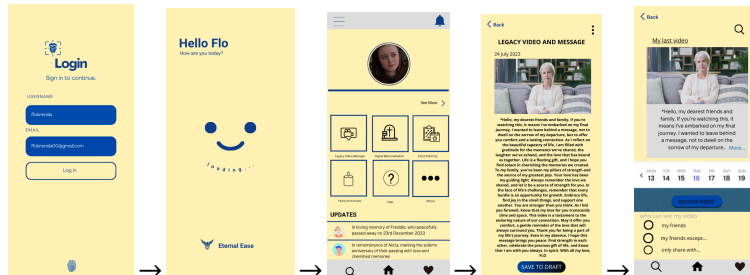
Login→Loading→Main menu→Event planning



Login→Loading→Main menu→Digital memorialization



Login→Loading→Main menu→Legacy video message



5.2 What Would Be Done Differently

It is essential to increase user engagement through in-depth interviews, surveys, and targeted group discussions in order to improve the design of our funeral app. We hope to capture a wide range of viewpoints and preferences by involving a more numerous and diverse group of respondents or testers. Comprehensive interviews yield qualitative understanding, questionnaires present numerical information, and targeted focus groups encourage conversational exchanges. This all-encompassing strategy guarantees a sophisticated comprehension of user expectations, directing well-informed and focused design choices. We hope to develop a funeral app that truly satisfies the wide range of needs of our user community by actively involving a diverse array of testers and guaranteeing relevance, usability, and caring support during significant life events.

5.3 Specific Design Decisions

User authentication

Include a multi-layered authentication system that combines biometric (facial recognition or fingerprint) authentication, Two Factor Authentication (2FA), and biometric authentication (fingerprint or facial recognition).

Profile management

The integration of Augmented Reality (AR) and Internet of Things (IoT) is transforming the Profile Management of our funeral app. By incorporating a dynamic visual layer, augmented reality (AR) enables users to strategically place digital things inside holographic environments. Concurrently, IoT guarantees smooth updates, easily adding new memories. Combining these technologies shows our dedication to a sophisticated and user-centric approach by maintaining an organized profile and customizing the experience to each individual's preferences.

Legacy video message

Our funeral app's Legacy Video Messages feature, which gives users an exclusive way to record and save sentimental moments for later, is the main attraction. With the ability to record video messages and plan their release for particular dates or occasions, this feature guarantees a legacy that is consistent with significant anniversaries. Our innovative design integrates AI analysis of user-shared text and media content in addition to allowing users to schedule the future distribution of their messages. This deliberate strategy, which includes both written notes and Legacy Video Messages, turns the feature into a moving instrument for creating emotionally charged connections and offering solace when thinking back on important events.

Digital memorialization

Digital memorialization, which provides a specific area for remembering loved ones digitally, is important in our funeral app. Users can easily share stories, movies, and images to make emotional digital memorials. A key design choice promotes harmony and preserves legacies by introducing a virtual memorial wall for group notes. Custom avatars in augmented reality (AR) add a personal touch and foster a sense of companionship. Collaborative AR experiences facilitate worldwide engagement, transforming remembrance into a communal and significant virtual experience.

Event planning

The Event Planning feature in our funeral app is a strategic tool that makes memorial event planning easier. Users work together to choose an appropriate time and date by using calendars, polls, and forms. Polls facilitate group decision-making, forms collect preferences, and calendars complete the schedule. Notifications and reminders improve engagement, and

AI-powered scheduling makes proactive recommendations for the best times. This meticulous planning guarantees a meaningful, well-attended event that reflects our dedication to considerate and inclusive memorial ceremonies.

Yearly anniversaries

Our funeral app's Yearly Anniversaries feature, which allows users to proactively honor loved ones with personalized reminders for important milestones, redefines remembrance. Users can virtually light candles and incense on the anniversary as a sign of respect. The commemoration is improved by sending virtual flowers and holding online memorial gatherings. Users who explore virtual reality (VR) encounter immersive memorial events. VR spaces that are integrated with event planning enable users to plan customized virtual memorials that span distances to create a cohesive and powerful remembrance.

5.4 Features to Prototype

Login

- Enables users to login using their username and password
- Enables new users to sign up to the app
- Enables fingerprint and facial recognition

Register

- Enables users to register their accounts using their name,email and password

User profile

- Allow user view feed
- Allow user to edit their profile
- Allow user to invite friends and share the app
- Include search option to search for other user

Event planning

- Consist of calendar to mark the upcoming event
- Provide the poster for each event
- Provide the hall with the time available
- Provide the easier booking confirmation

Yearly anniversaries

- Includes flowers column where the guest for the anniversary will give virtual flower to the deceased ones
- Commemoration box is allow the guests to celebrate the deceased ones by writing messages about them
- Meeting link is provided to join the yearly anniversary virtually
- Includes VR session of the deceased ones

Legacy video messages

- Synchronize calendar with the time of the video posting
- Provide edit video and messages
- Provide the post privacy

5.5 Prototype Techniques

With Canva for the final design, our funeral app prototype technique smoothly combines low-fidelity prototyping through early sketches with a transition to higher-fidelity representation. We quickly and iteratively sketch ideas on paper to get the design process started, which enables us to quickly explore a variety of concepts and user flows. This low-fidelity method makes feedback collection and early usability testing easier. We use Canva to refine visual elements as the design develops, adding polished details, color schemes, and overall aesthetics. Moving to Canva signifies a step towards higher-fidelity design and guarantees a polished and aesthetically pleasing finished product. User feedback from the Canva designs and sketches is crucial to this iterative process because it enables us to gradually improve the funeral app's functionality, design, and user experience while eventually aiming for a compassionate and user-friendly digital platform.

5.7 Member Contributions

NAME	CONTRIBUTION
LEENA A/P SELLAMANY	1.0 introduction and slide
SHWEHTTA PARANI KUMAR	2.0 Final Design overview
SYAZA SYAZANA BINTI ANWAR	3.0 Implementation
WAN NURADRIANA BALQIS BT WAN AZMI	4.0 Evaluation results
NORSYLAH BINTI SAFRUDIN	5.0 Reflection and slide

5.8 Conclusion

In conclusion, the funeral app positions itself as a heartfelt and significant digital ecosystem that has been thoughtfully created to navigate the challenging landscape of memorialization. The application creates a sense of security and personalisation by allowing users to choose and save the memories of their ancestors. It is reinforced by features like robust user authentication and profile management. Including legacy video messages not only makes proactive planning possible, but it also makes remembering a lasting experience. The digital memorialization feature, with its elements of augmented and virtual reality, creates a sacred space where people can cherish and share their most meaningful memories. Meanwhile, the state-of-the-art yearly anniversaries feature eliminates barriers and fosters harmony and support with its virtual gestures and immersive experiences. Collectively, these thoughtful features create a story of empathy and comprehension, positioning the funeral app not just as a technological tool, but as a heartfelt companion on the journey of remembrance, providing solace and connection in the digital tapestry of life and loss.

6.0 References

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